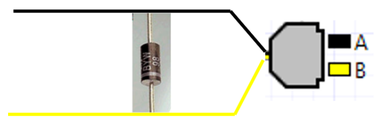
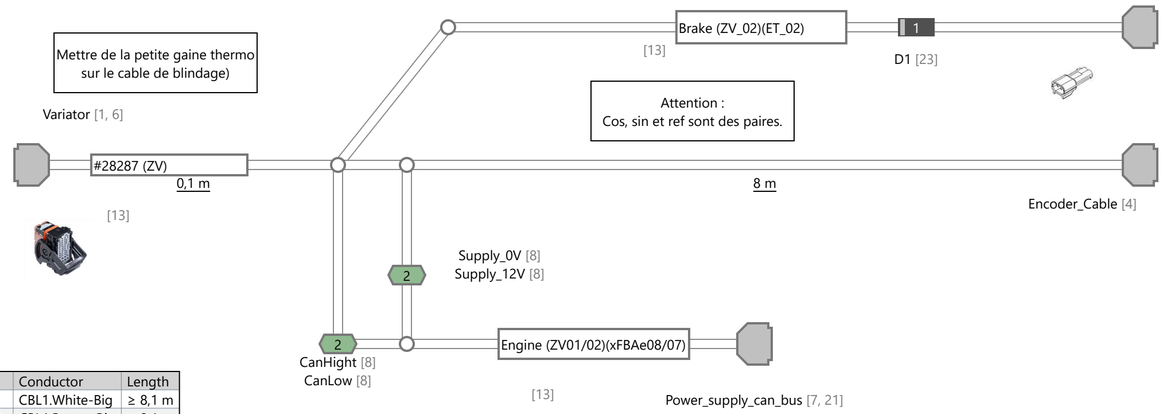
Revisions			
Rev.	Date	Author	Description
A	18/09/2019	Florent de Labarre	
B	26/11/2019	Florent de Labarre	Double Can
C	20/12/2019	Florent de Labarre	Mise à jour frein
D	17/03/2020	Florent de Labarre	Ajout diode de roue libre
E	10/07/2020	Florent de Labarre	Rectification label engine

From	To	Conductor	Color	Gauge
D1.NEG	Brake_M.B	W17.Yellow		1 mm²
D1.NEG	Variator.K3	W14.Yellow		1 mm²
D1.POS	Brake_M.A	W20.Gray-Red		1 mm²
D1.POS	Power_supply_can_bus.E	W12.Violet		1 mm²





Brake_M

Cavity	To	Conductor	Color	Gauge	From Contact PN
A	D1.POS	W20.Gray-Red		1 mm²	
B	D1.NEG	W17.Yellow		1 mm²	

Encoder_Cable

Cavity	To	Color	Gauge	Conductor	Length
1_ref_high	Variator.G2		0,25 mm²	CBL1.Pink	≥ 8,1 m
2_ref_low	Variator.G1		0,25 mm²	CBL1.Gray	≥ 8,1 m
3_cos_high	Variator.F4		0,25 mm²	CBL1.Red	≥ 8,1 m
4_cos_low	Variator.F3		0,25 mm²	CBL1.Blue	≥ 8,1 m
5_sin_high	Variator.F2		0,25 mm²	CBL1.Brown	≥ 8,1 m
6_sin_low	Variator.F1		0,25 mm²	CBL1.Green	≥ 8,1 m
9_KTY_high	Variator.A4		0,5 mm²	CBL1.Brown-Big	≥ 8,1 m
10_KTY_low	Variator.A3		0,5 mm²	CBL1.White-Big	≥ 8,1 m
Shield	Supply_0V				≥ 8 m

Variator

Cavity	To	Color	Gauge	Conductor	Length
A3	Encoder_Cable.10_KTY_low		0,5 mm²	CBL1.White-Big	≥ 8,1 m
A4	Encoder_Cable.9_KTY_high		0,5 mm²	CBL1.Brown-Big	≥ 8,1 m
B2	CanHight		0,5 mm²	W11.White	≥ 0,1 m
B3	CanLow		0,5 mm²	W15.Blue	≥ 0,1 m
C2	CanHight		0,5 mm²	W7.White	≥ 0,1 m
C3	CanLow		0,5 mm²	W6.Blue	≥ 0,1 m
F1	Encoder_Cable.6_sin_low		0,25 mm²	CBL1.Green	≥ 8,1 m
F2	Encoder_Cable.5_sin_high		0,25 mm²	CBL1.Brown	≥ 8,1 m
F3	Encoder_Cable.4_cos_low		0,25 mm²	CBL1.Blue	≥ 8,1 m
F4	Encoder_Cable.3_cos_high		0,25 mm²	CBL1.Red	≥ 8,1 m
G1	Encoder_Cable.2_ref_low		0,25 mm²	CBL1.Gray	≥ 8,1 m
G2	Encoder_Cable.1_ref_high		0,25 mm²	CBL1.Pink	≥ 8,1 m
K3	D1.NEG		1 mm²	W14.Yellow	≥ 0,1 m
L1	Supply_12V		0,5 mm²	W8.Red	≥ 0,1 m
L4	Supply_0V		0,5 mm²	W13.Black	≥ 0,1 m
M1	Supply_12V		0,5 mm²	W10.Red	≥ 0,1 m
M4	Supply_0V		0,5 mm²	W3.Black	≥ 0,1 m

Power_supply_can_bus

Cavity	To	Conductor	Color	Gauge	From Contact PN	Twisted With
A	Supply_12V	W2.Red		1 mm²	Delphi_150_contact_male	
B	Supply_0V	W1.Black		1 mm²	Delphi_150_contact_male	
C	CanHight	W5.White		1 mm²	Delphi_150_contact_male	
D	CanLow	W16.Blue		1 mm²	Delphi_150_contact_male	
E	D1.POS	W12.Violet		1 mm²		

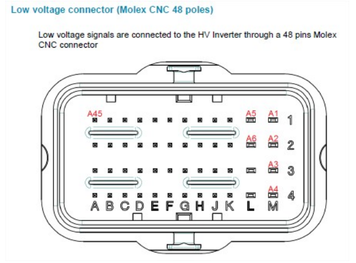
Supply_0V, Supply_12V

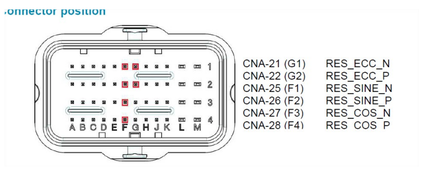
From	To	Conductor	Color	Gauge
Supply_0V	Encoder_Cable.Shield			
Supply_0V	Power_supply_can_bus.B	W1.Black		1 mm²
Supply_0V	Variator.L4	W13.Black		0,5 mm²
Supply_0V	Variator.M4	W3.Black		0,5 mm²
Supply_12V	Power_supply_can_bus.A	W2.Red		1 mm²
Supply_12V	Variator.L1	W8.Red		0,5 mm²
Supply_12V	Variator.M1	W10.Red		0,5 mm²

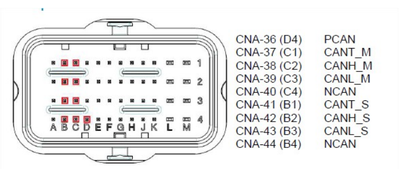
Codeur Check Mesurement, test before start :
Ref (Pink/ Grey) : 80 Ohm
Cos (Red/ Blue) : 70 Ohm
Sin (Yellow-Brown/ Green) : 70 Ohm
Brake : quelque ohm
KTY (PT1000): 500 Ohm
CAN BUS (A4-D4) : 60 Ohm

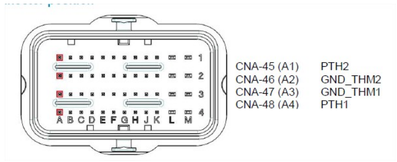
CanHight, CanLow

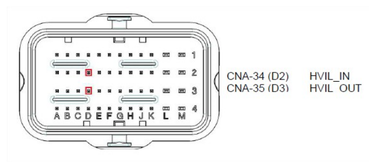
From	To	Conductor	Color	Gauge
CanHight	Power_supply_can_bus.C	W5.White		1 mm²
CanHight	Variator.B2	W11.White		0,5 mm²
CanHight	Variator.C2	W7.White		0,5 mm²
CanLow	Power_supply_can_bus.D	W16.Blue		1 mm²
CanLow	Variator.B3	W15.Blue		0,5 mm²
CanLow	Variator.C3	W6.Blue		0,5 mm²











Pin	Coor.	Type	Name	Description
A1	M1	Input	KEY	Input of the key switch signal (low voltage supply positive 12V or 24V)
A2	M2	Input	PEV	Connect the positive supply of electrovalves (EB, EVP and EVx) to this pin. Take the positive supply immediately after the main contactor.
A3	M3	Output	PANIN	Positive supply for external analog device (Sin/Cos sensor, potentiometers, ecc) factory selectable +12/+5V, maximum 100mA
A4	M4	Input	N_KEY	Low voltage power supply negative
A5	L1	Input	KEY	Input of the key switch signal (low voltage supply positive 12V or 24V)
A6	L2	Input	PEV	Connect the positive supply of electrovalves (EB, EVP and EVx) to this pin. Take the positive supply immediately after the main contactor.
A7	L3	Output	AGND	Ground for external analog devices (Sin/Cos sensor, potentiometers, ecc)
A8	L4	Input	N_KEY	Low voltage power supply negative

Bill of Materials

Id	Type	Manufacturer	Part Number	Description	Quantity
1	Boot	Molex	Connecteurs automobile CMC WIRE CAP FOR 28P AND 48P	538-64320-1301	1
2	Cable		2 Wire (1 twisted pair), 20 AWG Unshielded	Generic Cable	0 in
3	Cable	LAPP	#20811 - Câble vert OLFLEX (4x2x0.25mm² + 2x0.5mm²) LAP0036927		≥ 8,1 m
4	Connector		CMP100_encodeur_cable		1
6	Connector	Molex	#27627- Connecteur Molex 48 Pin	538-64320-1311	1
7	Contact	Delphi	Delphi_150_contact_male		4
8	Splice		Generic Splice	Generic Splice	4
9	Wire		0.5mm² Black		≥ 0,2 m
10	Wire		0.5mm² Red		≥ 0,2 m
11	Wire		1mm² Black		≥ 0 in
12	Wire		1mm² Red		
13	Label		Generic White Label	Generic Label	3
14	Wire		0.5 mm² - Blue		≥ 0,2 m
15	Wire		0.5 mm² - White		≥ 0,2 m
16	Wire		1 mm² Violet	Generic Wire	
17	Connector	Delphi	#16777 - Delphi_280_2_way_male	829-15300002	1
18	Wire		1 mm² Yellow	Generic Wire	≥ 0,1 m
19	Wire		1mm² - White		≥ 0 in
20	Wire		1mm² - Blue		
21	Connector		#14017 - Delphi_150_6_way_male	829-12124107	1
22	Wire		0.75mm2 yellow		0 in
23	Diode		Diode traversante, 5A, 60V, SB560-E3/54	#29732	1
24	Wire		Grey - red stripe 1 mm2		

	Name	Date
Drawn By	----	----
Eng Approved	----	----
Mfg Approved	----	----
QA	----	----

Title: #28287 - Engine_zapi_encoder_controler (ZV) - A2

Description

Revision: ----

This document contains confidential and proprietary information that cannot be reproduced or divulged, in whole or in part, without written authorization.